PYTHON PROGRAMMING



21 Days Challenge

Day-16

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BANK MANAGEMENT SYSTEM USING OOP



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Abstract Class (Abstraction)

```
from abc import ABC, abstractmethod   # For Abstraction


class Account(ABC):
    def __init__(self, acc_no, name, balance):
        self.acc_no = acc_no          # Public Attribute
        self._name = name             # Protected Attribute
        self.__balance = balance      # Private Attribute
```

Public Methods

Public method to show Account No

```
def show_Account_No(self):
    return f"Account No: {self.acc_no}"
```

Setter method

Setter method to update balance safely

```
def deposit(self, amount):
        if amount > 0:
            self.__balance += amount
            print(f"{amount} Tk deposited successfully!")
        else:
            print("Invalid deposit amount!")
```

Getter Methods

Private attribute access with getter

```
def get_balance(self):
        return self.__balance
```

Abstract method

Abstract method must be implemented by subclasses

```
@abstractmethod
def withdraw(self, amount):
    pass
```

Inheritance & Polymorphism

```
class SavingsAccount(Account):
    def withdraw(self, amount):
        if amount <= self.get_balance():
            print(f"{amount} Tk withdrawn from Savings Account")
        else:
            print("Insufficient balance in Savings Account!")
```

Inheritance & Polymorphism

```
class CurrentAccount(Account):
    def withdraw(self, amount):
        if amount <= self.get_balance() + 5000:  # Overdraft limit
            print(f"{amount} Tk withdrawn from Current Account
        else:
            print("Overdraft limit exceeded!")
```

Object Creation

```
savings = SavingsAccount(101, "Abdullah", 10000)
current = CurrentAccount(102, "Rahman", 5000)
```

Using Methods

```
print("\n--- Savings Account ---")
print(savings.show_balance())
savings.deposit(2000)
print(savings.show_balance())
savings.withdraw(3000)
```

Using Methods

```
print("\n--- Current Account ---")
print(current.show_balance())
current.withdraw(8000)   # Allowed due to overdraft
current.withdraw(15000)  # Exceeds overdraft
```

Polymorphism Example

```
print("\n--- Polymorphism Demo ---")
accounts = [savings, current]

for acc in accounts:
    acc.withdraw(2000)  # Same method name behaves differently
```



ASSIGNMENT

Design a simple OOP-based project Library Management System that demonstrates the core concepts of object-oriented programming. Your project should include at least one abstract class, subclasses using inheritance, attributes with different access modifiers (public, protected, private), and methods that show polymorphism by behaving differently in different subclasses. Finally, create objects to test the system and show how abstraction, inheritance, polymorphism, and encapsulation work together in a real-world scenario.

QUESTION SESSION

THANK YOU

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